

VIJAYA NEEHARIKA RANGISETTY

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TECHNICAL SKILLS

Frontend Technologies: ReactJS, Angular, Vue, Bootstrap, HTML, CSS, Tailwind CSS, Streamlit

Languages Frameworks: C++, Java, JavaScript, TypeScript, Node.js, Python, Spring Boot, Express.js

Cloud DevOps: AWS (Amplify, Lambda, Bedrock), Jenkins, GitHub, GitLab CI/CD, Docker, YAML scripting

Services Architecture: REST APIs, Microservices, Distributed Systems, Load Balancing, Fault Tolerance, RAG Pipelines

Databases: PostgreSQL, MongoDB, MySQL, Microsoft SQL Server

Tools: JIRA, Confluence, Postman, Swagger, Figma, Web Inspector

EXPERIENCE

People Tech Group

February 2024 - Present

Software Engineer

Seattle, Washington

- Developed a TestRail MCP integration and implemented the client-side code for seamless Strands and NLP Strands integration, enhancing automated test management, reporting, and natural language-driven workflows.
- Built a CloudWatch Model Context Protocol (MCP) server to analyze logs in real time using Bedrock LLMs, enabling scalable alerting, anomaly summarization, and observability enhancements across AWS environments.
- Leading the backend and infrastructure layer of an AI-based compliance and observability suite for aerospace and automotive domains, integrating **Claude LLMs**, real-time dashboards, and **semantic search** pipelines.
- Designed and implemented a **ticket validator** for General Motors that automates intake enforcement, using OpenSearch embeddings, and uses CLAUDE for evidence gap identification—reducing manual triage time by over 60%.
- Designed the technical architecture diagram for General Motors, mapping all key system components, data pipelines, and service interactions for scalable deployment.
- Developed and deployed a Retrieval-Augmented Generation (**RAG**) pipeline leveraging **AWS Bedrock** and **OpenSearch** to dynamically generate traceability matrices for DO-178C compliance audits.
- Built distributed system simulators in **C** for real-time flight control modeling and integrated them with cloud-hosted FastAPI endpoints—achieving seamless data interchange between native code and web services.
- Delivered an interactive observability dashboard in **React** and **Tailwind CSS** to visualize AI model metrics (accuracy, latency, drift), improving LLM regression identification time by 3x.
- Designed and maintained **GitLab CI/CD** pipelines with YAML to automate builds, testing, and deployments, improving release reliability and supporting compliance requirements for aerospace and automotive projects.
- Developed Python-based automation scripts to support test orchestration, validation, and reporting across embedded and cloud-native systems.
- Integrated automated unit and regression testing into CI/CD workflows using Python, reducing manual QA effort and accelerating validation cycles for safety-critical software.
- Conducted peer code reviews and participated in sprint planning and design discussions for scalable backend services.

Pilomation

June 2024 - December 2024

Software Engineer

Fairfax, Virginia

- Contributed to the design and development of an AI-powered film production management platform that optimized script planning, actor scheduling, crew coordination, and shoot logistics.
- Engineered scalable backend services in **FastAPI** and **SQL** to manage real-time data relationships between scripts, talent profiles, production assets, and calendar events.
- Developed modular, mobile-responsive UIs in **React** from **Figma** prototypes, supporting complex user flows for booking, versioning, and live updates across departments.
- Introduced role-based access control (**RBAC**) for actors, producers, and admins, ensuring granular permissions and auditability of project timelines and documents.

- Collaborated with PMs, designers, and QA in agile sprints to ship features across production analytics, asset management, and real-time notifications.

Teleparadigm Network Ltd

Software Engineer

December 2020 – May 2022

Hyderabad, Telangana

- Developed a full-stack student and academic management portal using **Node.js, Express.js, MongoDB, and React** to enable real-time registration, attendance, and communication for students and faculty.
- Designed and implemented **REST APIs** for secure user authentication, academic performance tracking, and notifications—supporting multi-role access and modular data control
- Created a **responsive frontend UI** that offered dynamic dashboards for administrators and personalized experiences for students—reducing onboarding issues by 40%.
- Contributed to secure deployment and scalability improvements, ensuring platform performance under concurrent user loads during registration cycles.
- Built unit and integration tests with Postman achieving 85%+ coverage.

PROJECTS

AI-Powered Healthcare Report Analysis System

- Developed a smart document parsing and triage tool using **Python, LangChain, and AWS Bedrock (Claude)** to extract and analyze unstructured data from lab reports, prescriptions, and discharge summaries
- Implemented an end-to-end pipeline that performs medical entity extraction, symptom-diagnosis mapping, and follow-up recommendation generation using LLMs.
- Enabled doctors and clinical staff to save over 50% time in manual report interpretation by converting documents into machine-readable structured formats.
- Deployed the solution as a Streamlit-based SPA, enabling clinicians to input PDFs or text data and receive structured summaries and red-flag alerts.

Real-Time User Sentiment Dashboard for Product Feedback

- Developed a live dashboard that analyzes and visualizes customer feedback sentiment from multiple platforms using Python, FastAPI, and WebSockets.
- Integrated real-time NLP pipelines using **Hugging Face** transformers to classify sentiment (positive/neutral/negative) and extract trending keywords and feature requests.
- Implemented **Kafka**-based ingestion and **Redis caching** to ensure low-latency updates and scalable message processing.

Blog Platform

- Built a modern blogging application with dark mode, rich-text editor, **Firestore** authentication, and role-based access for admins and readers.
- Enabled image uploads, tag-based filtering, and real-time search using MongoDB query optimization and React dynamic routing.
- Implemented content moderation features like profanity filtering and spam detection using cosine similarity and simple NLP techniques.
- Deployed using NGINX and GitHub Actions CI/CD pipeline for consistent updates and performance tracking.

EDUCATION

George Mason University

Masters in Computer Science

May 2024

Fairfax, Virginia

Neil Gogte Institute of Technology

Bachelors in Engineering

June 2022

Hyderabad, Telangana

CERTIFICATIONS

AWS Certified Associate Developer, AWS

AWS Certified Cloud Practitioner, Amazon Web Services

Google UX Design, Google

Full Stack Developer, Coursera